"LOW-POWER IOT-BASED BRIGHTNESS SENSING AND LED CONTROL USING TELOSB AND CONTIKI"

OUR CODE-YOUR INNOVATION

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OBJECTIVE:

To design and implement a wireless light intensity monitoring system using Contiki-NG, where a sender node (Mote 1) transmits light sensor data and a receiver node (Mote 2) displays the received data and adjusts LED indicators accordingly.

METHODOLOGY:

- Hardware & Software Setup
  Hardware: Sky Motes (or similar Contiki-compatible devices)
  Software: Contiki-NG, C programming, UDP communication
- Implementation Steps
  Sender (Mote 1) – Light Sensor Node
  - Activates button and light sensor.
  - On button press, reads light intensity value.
  - Sends light intensity data via UDP to the receiver node.
- Receiver (Mote 2) – Display & LED Indication
  - Waits for button press to activate data reception.
  - Receives UDP packets containing light intensity data.
  -Processes the received data and adjusts LEDs based on intensity.
  - High intensity → Red LED ON
  - Medium intensity → Green LED ON
  - Low intensity → Blue LED ON
  - LEDs remain ON until the next button press.

Applications

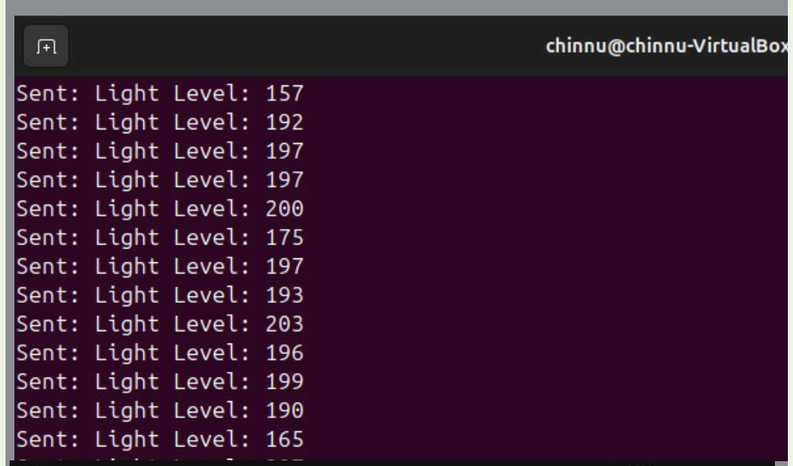
- Smart lighting systems
- Environmental monitoring
- Wireless sensor networks
- IoT-based home automation

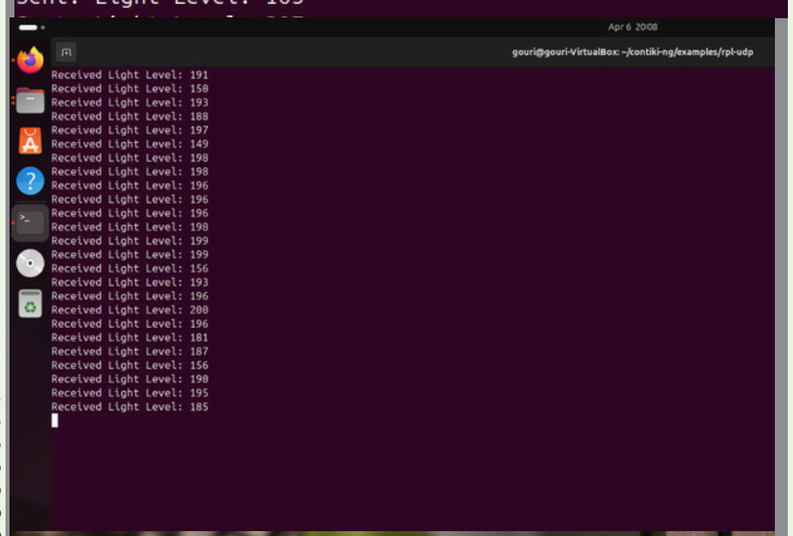
Analysis

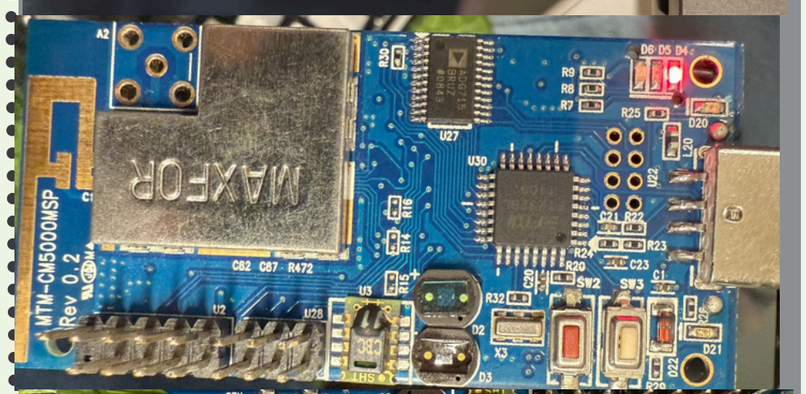
- Ensured real-time light monitoring via UDP communication.
- Optimized low-power wireless communication in Contiki-NG.
- Demonstrated event-driven execution for efficient processing.

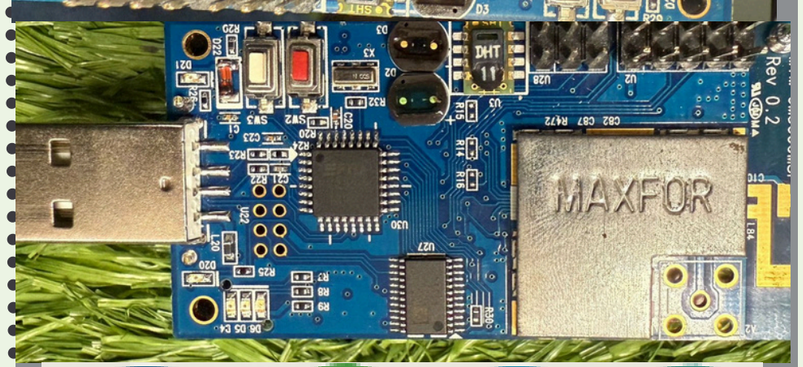
Conclusion: The implemented system successfully transmits and processes light intensity data using UDP over a wireless sensor network. The receiver node dynamically adjusts LED indicators based on real-time sensor readings, showcasing an effective IoT-based solution for smart environments.

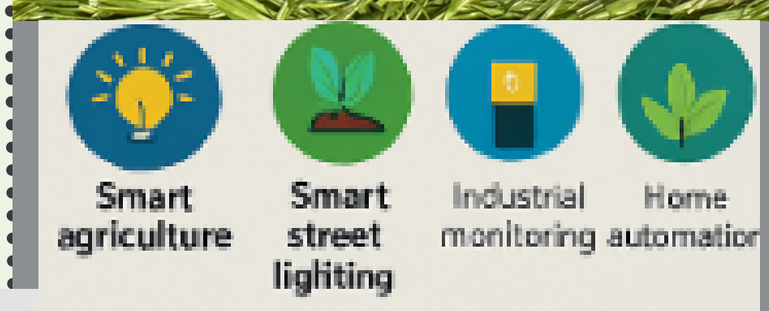
RESULTS











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